

THE MIND GENESIS ARCHITECTURE (TMGA)

MULTI-AGENT REFERENCE TESTBED

Synthetic Evaluation of Governance Arbitration, Drift Monitoring, Containment Logic, and Architectural Failure Modes

Supporting Research Documentation — 2026

1. Purpose and Scope

This document describes the multi-agent reference testbed accompanying *The Mind Genesis Architecture: A Multidimensional Model for Shared Intelligence Ecosystems*.

The testbed is designed to examine selected TMGA governance mechanisms under synthetic and reproducible conditions. Its purpose is not to establish empirical validation of the complete TMGA architecture, demonstrate real-world safety, or reproduce the full 33-dimensional taxonomy.

Instead, the simulations provide a controlled environment in which selected architectural predicates can be executed, observed, compared, and subjected to deliberately constructed failure conditions.

The testbed focuses primarily on:

- governance arbitration;
- drift monitoring;
- epistemic restraint;
- Circuit Breaker Protocol (CBP) activation;
- bounded operational degradation;
- adversarial and noisy measurement conditions;
- multi-agent disagreement;
- comparative activation strategies; and
- selected interactions among seven critical governance components.

The results should therefore be interpreted as **simulation-level behavioral observations under specified synthetic conditions**.

2. Architectural Context

TMGA proposes 33 dimensions distributed across three concurrent and non-hierarchical spheres:

AGI — Cognitive Breadth

ASI — Capability Scale

AWI — Meaning and Responsible Restraint

The 33 dimensions constitute a proposed architectural taxonomy and design vocabulary. They are not claimed to be empirically verified components of cognition or a mathematically necessary decomposition of intelligent systems.

The reference simulations do not instantiate all 33 dimensions as independently modeled cognitive processes.

Instead, they isolate selected governance mechanisms so that their behavior can be inspected under controlled conditions.

The central architectural principle evaluated by the testbed is:

Capability may generate a proposed action, while verification, constraint, and governance determine whether that action is authorized to proceed.

3. Reference Testbed Architecture

The simulation environment represents selected TMGA mechanisms through synthetic state vectors, confidence values, drift measurements, weighted arbitration, and containment predicates.

The principal simulated elements include:

Component	Reference-Testbed Role
NHCP	Architectural arbitration among heterogeneous recommendations and constraints
AWI Void	Represents epistemic restraint when modeled ethical confidence falls below a threshold
AWI Sentry	Represents monitoring of governance-relevant drift
AGI Guard	Represents bounded execution constraints
ASI Guard	Represents high-scale constraint boundaries
ASI Verify	Represents verification conditions within the extended simulation
CBP	Represents temporary containment following specified persistent-drift conditions
Bounded Degradation / Triage	Represents selective reduction of modeled capabilities under stress

NHCP in this testbed should not be interpreted as a replacement for distributed consensus protocols such as Paxos or PBFT. It represents architectural arbitration among heterogeneous recommendations and constraints rather than low-level agreement among distributed nodes.

4. Simulation Design

The accompanying repository contains four reference simulation scripts.

Script	Manuscript Section	Evaluation
<code>nhcp_simulation.py</code>	Section 8.5	Base reference scenarios
<code>nhcp_simulation_advanced.py</code>	Section 8.6	Adversarial, noisy, multi-agent, and ablation scenarios
<code>nhcp_simulation_comparative.py</code>	Sections 8.7–8.8	Comparative activation strategies and threshold sensitivity
<code>nhcp_simulation_extended.py</code>	Section 8.9	Seven-component extended simulation

Each reported scenario is evaluated over 100 runs under the conditions defined by the corresponding script.

Fixed random seeds are used where stochastic behavior is introduced, supporting reproducibility of the reported numerical outcomes under the specified reference conditions.

5. Base Reference Scenarios

The base simulation evaluates three deliberately simplified governance conditions.

Scenario	Runs	Reference Outcome
NOMINAL	100	100 EXECUTED
STRESS	100	100 SUSPENDED
CBP	100	100 CBP_CONTAINMENT

These scenarios are sanity checks for the implemented predicates.

The **NOMINAL** scenario verifies that the modeled arbitration pathway permits execution under the specified low-risk conditions.

The **STRESS** scenario verifies that the modeled confidence threshold can produce suspension under the specified conditions.

The **CBP** scenario verifies that the implemented persistent-drift predicate can enter the modeled containment state when its trigger conditions are satisfied.

These results demonstrate the behavior of the reference implementation under its own predefined conditions. They do not establish external validity or real-world safety performance.

6. Advanced Synthetic Scenarios

The advanced simulation introduces three more demanding measurement environments.

6.1 Adversarial Drift Measurement

The ADVERSARIAL scenario deliberately manipulates the drift measurement before it reaches the CBP predicate.

Under the specified reference conditions, the simulation produces:

0 / 100 CBP containment outcomes.

This result exposes a dependency of the modeled containment mechanism on the integrity of its measurement channel.

It should not be interpreted as demonstrating adversarial robustness. On the contrary, the scenario identifies a failure mode in which manipulation of the input signal prevents the containment predicate from receiving sufficient evidence for activation.

The result motivates future investigation of measurement integrity, independent sensing, cryptographic attestation, and hardware-rooted trust mechanisms.

Such countermeasures remain proposed design directions rather than validated implementations.

6.2 Noisy Measurement

The NOISY scenario introduces stochastic variation into the drift measurement.

Under the specified conditions:

53 / 100 runs enter CBP containment.

The remaining runs do not satisfy the complete CBP activation predicate.

This result illustrates the sensitivity of the reference predicate to measurement noise. It does not establish an empirically calibrated false-positive or false-negative rate because the synthetic noise model is not derived from deployment data.

6.3 Multi-Agent Disagreement

The MULTI-AGENT scenario generates multiple drift reports and uses median aggregation to represent disagreement among measurement sources.

Under the specified conditions:

56 / 100 runs enter CBP containment.

The result illustrates how the modeled activation predicate behaves when its input is derived from heterogeneous reports rather than a single deterministic measurement.

It does not demonstrate that median aggregation is sufficient for real multi-agent governance, Byzantine robustness, or adversarial consensus.

7. Ablation Study

The advanced simulation also includes a simplified ablation study involving CBP, Triage, and a modeled NMI-related flag.

The ablation is intended to distinguish logical paths in the reference implementation.

Configuration	Reference Result
FULL_ARCHITECTURE	100 / 100 CBP_CONTAINMENT
NO_CBP	100 / 100 EXECUTED
NO_TRIAGE	100 / 100 CBP_CONTAINMENT
NO_NMI	100 / 100 CBP_CONTAINMENT

Within this synthetic implementation, removing CBP changes the containment outcome, while removing the modeled Triage or NMI-related flags does not alter CBP activation under the tested condition.

This should be interpreted narrowly.

The ablation demonstrates separation among specific logical paths encoded in the simulation. It does **not** experimentally establish the effectiveness of a physical NMI mechanism, hardware containment, actuator isolation, or deployment-level emergency control.

8. Comparative Activation Strategies

The comparative simulation evaluates three CBP activation strategies:

TMGA predicate: three consecutive threshold breaches together with a positive drift derivative.

Single-breach baseline: activation following any individual threshold breach.

Three-cycle average baseline: activation when the average of three measurements exceeds the specified threshold.

The reference results are:

Scenario	TMGA Predicate	Single-Breach	Three-Cycle Average
ADVERSARIAL	0 / 100	21 / 100	50 / 100
NOISY	53 / 100	100 / 100	100 / 100
MULTI-AGENT	56 / 100	99 / 100	100 / 100

These comparisons characterize different activation behaviors under the same synthetic scenarios.

They do not establish that one strategy is universally safer or superior. The relative desirability of activation sensitivity, delayed intervention, false positives, and missed containment would depend on deployment-specific costs and calibrated risk models.

9. Threshold Sensitivity

The sensitivity analysis varies the CBP drift threshold while holding the reference scenario definitions otherwise constant.

τ_{drift}	NOISY	MULTI-AGENT
0.03	53 / 100	57 / 100
0.05	53 / 100	56 / 100
0.07	52 / 100	44 / 100
0.10	30 / 100	7 / 100
0.15	0 / 100	0 / 100

The results show that containment frequency in the reference implementation depends materially on the selected threshold.

Accordingly, $\tau_{\text{drift}} = 0.05$ should be understood as an illustrative simulation parameter rather than an empirically established operational threshold.

Deployment would require domain-specific calibration and evaluation of the consequences of both unnecessary activation and missed activation.

10. Extended Seven-Component Simulation

The extended simulation activates seven critical governance components simultaneously:

1. AGI Guard
2. ASI Guard
3. ASI Verify
4. AWI Void
5. AWI Sentry
6. CBP
7. NHCP

The reference outcomes are:

Scenario	Reference Outcome
NOMINAL	100 / 100 EXECUTED
STRESS	100 / 100 SUSPENDED
CBP	100 / 100 CBP_CONTAINMENT
ADVERSARIAL	100 / 100 EXECUTED
NOISY	83 / 100 EXECUTED; 17 / 100 CBP_CONTAINMENT
MULTI-AGENT	83 / 100 EXECUTED; 17 / 100 CBP_CONTAINMENT

This experiment increases interaction complexity relative to the earlier reference scenarios.

It does not constitute a simulation of the entire 33-dimensional TMGA architecture.

The remaining 26 dimensions are not separately instantiated or empirically validated by this experiment.

11. Hardware Containment: Proposed Deployment Layer

TMGA discusses hardware-assisted containment as a possible future deployment mechanism.

Examples include:

- hardware-level interruption mechanisms;
- Non-Maskable Interrupts where supported by the target platform;
- command escrow or buffering;
- independent actuator interlocks;
- hardware-rooted measurement;
- cryptographic attestation; and
- protected audit pathways.

These mechanisms are **not physically instantiated in the reference testbed**.

Accordingly, terms such as “NMI” appearing in simulation variables or ablation labels represent architectural or logical conditions within the reference model. They should not be interpreted as evidence that an actual processor interrupt, physical actuator cutoff, hardware escrow queue, or measured hardware response occurred.

Any implementation of these mechanisms would require platform-specific engineering and independent experimental evaluation.

Timing examples discussed elsewhere in TMGA documentation should likewise be treated as illustrative architectural parameters rather than demonstrated performance guarantees.

12. What the Reference Testbed Supports

The testbed supports several limited conclusions.

It shows that selected TMGA governance predicates can be expressed computationally.

It shows that their behavior can be reproduced under specified synthetic conditions.

It allows specific failure modes to be exposed rather than hidden.

It allows alternative activation predicates and thresholds to be compared under common reference scenarios.

It provides an executable bridge between the conceptual architecture and future empirical investigation.

These are useful properties of a reference implementation, but they are not equivalent to validating TMGA as a complete architecture.

13. What the Reference Testbed Does Not Establish

The simulations do not establish:

- empirical validity of the 33-dimensional taxonomy;
- safety of an AGI, ASI, or AWI system;
- adversarial robustness in deployed environments;
- superiority of TMGA over alternative governance architectures;
- production readiness;
- regulatory conformity;
- physical hardware containment performance;
- effectiveness of a real NMI or actuator isolation mechanism;

- moral or ethical correctness of machine decisions;
- successful alignment of advanced autonomous systems; or
- guaranteed prevention of harmful actions.

The experiments should therefore be interpreted as **reference simulations for architectural analysis and falsifiable future research**, not as a safety certification.

14. Future Evaluation

The next stage of evaluation would require moving beyond synthetic predicates toward independently testable implementations.

Relevant research directions include:

Measurement calibration. Drift and confidence metrics should be evaluated against domain-specific empirical data.

Adversarial evaluation. Measurement channels, arbitration logic, audit mechanisms, and containment predicates should be subjected to systematic attack models.

Hardware prototyping. Proposed interruption, escrow, and actuator-isolation mechanisms should be implemented and benchmarked on representative hardware.

False-positive and false-negative analysis. Containment thresholds should be evaluated against independently specified ground truth and explicit intervention costs.

Human governance evaluation. Authorization, escalation, suspension, and recovery procedures should be studied with human and institutional decision-makers.

Fuller architectural simulation. Additional TMGA dimensions should be operationalized independently rather than inferred from the behavior of the seven-component reference model.

15. Reproducibility

The accompanying simulation code and stored numerical results are available in the TMGA repository.

The repository includes:

```
nhcp_simulation.py
nhcp_simulation_advanced.py
nhcp_simulation_comparative.py
nhcp_simulation_extended.py
```

and:

advanced_simulation_results.json
comparative_sensitivity_results.json
extended_simulation_results.json

The simulation code should be treated as a research reference implementation.

Reproduction of the reported numerical results demonstrates reproducibility of the specified computational experiment; it does not independently validate the architectural assumptions encoded by that experiment.

16. Document Status

Supporting research documentation for The Mind Genesis Architecture (TMGA), 2026.

This document accompanies:

The Mind Genesis Architecture: A Multidimensional Model for Shared Intelligence Ecosystems.

Repository:

<https://github.com/ahmett76/the-mind-genesis-architecture>